Predictive Thermal Battery Management System for Li-ion Battery Packs

Abstract

Thermal runaway is a major safety hazard in lithium-ion battery packs, and one of its main causes is an internal short circuit, which in its early stage slowly discharges the affected cell and generates only a fraction of a watt of heat. Conventional battery management systems (BMS) and battery protection ICs act only when a voltage, current or temperature crosses a fixed threshold, and therefore cannot detect this stage. This project presents a predictive thermal battery management system for a 4S1P lithium-ion pack of NCR18650GA cells. The system has two independent protection layers: a hardware BMS that provides over-voltage, under-voltage, over-current and short-circuit protection, and an ESP32-based supervisory controller that samples the four cell voltages through a 16-bit ADS1115 ADC, the pack current through an INA226 with a 5 mΩ shunt, and the four cell temperatures and the ambient temperature through NTC thermistors at 5 Hz. A two-state (core and surface) lumped thermal model of each cell is combined with a Kalman filter to estimate the core temperature and the abnormal heat generation, in watts, that the cell's I^2R losses cannot explain. The model predicts each cell's temperature 1 and 5 minutes ahead and confirms every prediction against the measured value. Additional algorithms track the internal resistance online, estimate the self-discharge current of each cell at rest from its relative voltage drift, cross-check the thermal and electrical evidence through an energy balance, forecast the time to the temperature limit and the safe continuous current, and detect self-accelerating heating. These indicators are combined into a four-level risk classification (Normal, Watch, Warning and Trip) that controls separate charge and load relays with latched, fail-safe tripping, and a Wi-Fi dashboard displays the estimates. Compared with the LSTM neural-network method of the base paper, which requires offline GPU training and responds only to rapid temperature rises of about 9 °C/min, the proposed model-based method needs no training data, runs on a low-cost microcontroller and is designed to detect abnormal heat of about 0.4 W (0.5 °C/min), a temperature rise about eighteen times slower. The system is evaluated through sensor calibration, controlled charge and discharge tests, and a safely emulated soft internal short circuit created with an external resistor.

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